

Ensemble benefit does not imply heterogeneity superiority

A Source ablation within the TG teacher

Grassmann-only
(G branch only)



Positive transfer

+0.0410 NLL

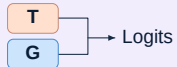
Transformer-only
(T branch only)



Positive transfer

+0.1349 NLL

Fused TG teacher
(T + G, $\alpha = 0.5$)



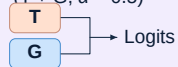
Positive transfer

+0.1557 NLL

Fused supervision transfers better than either component branch alone.

B Homogeneous control

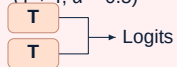
Heterogeneous TG
ensemble
(T + G, $\alpha = 0.5$)



Positive transfer

+0.1557 NLL

Homogeneous TT
ensemble
(T + T, $\alpha = 0.5$)



Positive transfer

+0.1681 NLL

TT yields lower student NLL than TG
for all three matched students.

Ensemble supervision helps, but neither architectural heterogeneity nor larger branch disagreement guarantees better transfer.